

Gokturk Colak

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EDUCATION

Istanbul University

Bachelor of Arts in Economics, English Program

- **GPA: 3.70/4.00**

Beyazit, Istanbul

Aug. 2022 – Present

University of Tübingen

Erasmus+ Exchange Program – Economics & Data Science

Tübingen, Germany

Apr. 2026 – Sep. 2026

- Applied Quantitative Methods in Economics & Finance (S310): completed assignments on hypothesis testing via Monte Carlo simulation, estimator bias/variance/MLE derivations, OLS and binary choice models (LPM, Logit), and time series analysis (stationarity, unit root tests, AR models) in R
- Introduction to Data Science (DS320): completed foundational tutorials in R and Python covering data manipulation and basic analysis

EXPERIENCE

Undergraduate Research Assistant

Istanbul University Informatics Department

Nov. 2023 – June 2024

Vezneciler, Istanbul

- Collected and organized various data sets
- Created complete database of Management Information Systems department heads across 81 Turkish provinces
- Used Microsoft Excel and Google Sheets for data management

PROJECTS

Algorithmic Trading System Development | *Python, AWS EC2, BinanceAPI*

Aug. 2025 - Dec. 2025

- Signal refinement using an XGBoost Classifier trained on RSI, MACD, and OBV indicators
- Backtesting for investment strategy validation and developing automated market analysis tools
- Hysteresis-based entry/exit logic to maintain momentum-driven positions and filter volatility

Academic Research Paper | *Word, Excel, EndNote, R*

Feb. 2025 – June 2025

- Conducted comprehensive 15-year macroeconomic analysis of the U.S. economy using official data sources and 20+ academic reports
- Analyzed key economic indicators including GDP, inflation, unemployment, public debt, and productivity trends
- Evaluated policy impacts across four major economic phases including Great Recession and COVID-19
- Produced structured academic report with detailed financial data tables and policy assessments

TEAMWORK & ACTIVITIES

Istanbul University Young Investors Club (Genç Yatırımcılar Kulübü)

Research Department Member

Istanbul, Turkey

Oct. 2025 – Present

- Conduct research and prepare draft analyses to support the Social Media Department's weekly insights and posts

Istanbul University Data Club

Vice President

Istanbul, Turkey

Oct. 2023 – Oct. 2024

- Organized an institutional visit to IBB Data Laboratory and managed corporate communications
- Volunteered at 22nd Academic Informatics Conference focusing on technology and innovation trends

High School T-MBA Council

President of Scientific and Artistic Projects

Hatay, Turkey

Sep. 2019 – June 2020

- Managed entrepreneurial project development and innovation initiatives
- Managed social responsibility project development and financing initiatives

TECHNICAL SKILLS

Programming Languages: Python, R

Tools: Excel, RStudio, Jupyter Notebook, Git, Github, gretl

Libraries: Pandas, NumPy, Matplotlib, SciPy, tidyverse

Machine Learning: Scikit-Learn, XGBoost, LightGBM, TensorFlow, Keras, caret, tidymodels

Languages: Turkish (Native), English (Advanced), French (Beginner)

CERTIFICATIONS

Machine Learning for Time Series Data in Python – DataCamp

Exploratory Analysis in R – DataCamp

Next Generation Banking Simulation – Isbank

Digital Banking – Isbank